RW-6596 **605101**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Chemistry (Week - End)

ORGANIC CHEMISTRY–I

(NCBCS-1995–2001 Batch)

Time : 3 Hours Maximum : 100 Marks

(5 × 20 = 100)

Answer **all** questions.

All questions carry equal marks.

1. (a) Write short notes on

(i) Mesomeric effect.

(ii) Steric inhibition of resonance.

(iii) Hydrogen bonding.

(20)

(*Or*)

(b) What are the kinetic and non kinetic methods for determining rate of organic reactions ?

(20)

2. (a) Short notes on

(i) Carbanions.

(ii) Carbenes and nitrenes.

(10)

(b) Explain the following.

(i) Microscopic reversibility.

(ii) The concept of aromaticity.

(10)

(*Or*)

(c) Explain the mechanism of Nitration, Nitrosation and Diazonium coupling.

(10)

(d) What is Friedel Craft reaction ? (10)

3. (a) What is Diels Alder reaction ? (8)

(b) What is Michael addition ? (6)

(c) What is Wittig reaction ? (6)

(*Or*)

(d) Short notes on :

(i) Perkin reaction.

(ii) Knovenagel reaction.

(iii) Mannich reaction.

(iv) Claissen ester condensation.

(20)

4. (a) Write the structure and synthesis of any three Heterocyclic compounds.

(20)

(*Or*)

(b) Write in detail about the biosynthesis of alkaloids.

(20)

5. (a) What are terpenes ? How they are classified ? (10)

(b) How camphor is synthesiszed ? (10)

(*Or*)

(c) Give the synthesis of Zingiberene, cadinene. (20)

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RW-6597 **605102**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Chemistry (Week–End)

INORGANIC CHEMISTRY—I

(NCBCS—1995–2001 Batch)

Time : 3 Hours Maximum : 100 Marks

(5 × 20 = 100)

Answer the following questions.

All questions carry equal marks.

1. (a) Write IUPAC names of the following Co-ordination compounds.

(i) $NH_4\left[Cr(NCS)_4(NH_3)_2\right]$.

(ii) $\left[Co(en)_2(ONO)Cl\right]Cl$. (3)

(*Or*)

(b) Draw the possible isomers of $\left[Co(en)Cl_2\right]^+$.

(3)

(c) Describe a method of determining the stability constant of a complex. (7)

(*Or*)

(d) Find out the magnetic moment of strong and weak octahedral field for Cr^{3+} and Fe^{2+} ions. (7)

(e) Explain Crystal field splitting in tetrahedral complexes. Account for the smaller value of crystal field splitting in tetrahedral than in octahedral complexes. (10)

(*Or*)

(f) (i) What are the postulates of Pauling's valence Bond theory ? (5)

(ii) Explain how magnetic criteria is useful to explain geometry of $\left[Mn(Br)_4\right]^{2-}$ complex ion. (5)

2. (a) Write a note on free electron theory of metallic bond.

(3)

(*Or*)

(b) What is the difference in spatial arrangement of carbon atoms in diamond and graphite.

(3)

(c) Discuss the crystal structures of rutile and ZnS.

(7)

(*Or*)

(d) What are semiconductors ? Explain the '*n*' and '*p*'-type semiconductors.

(7)

(e) Write a note on intermetallic compounds.

(10)

(*Or*)

(f) How are the unit cell dimensions determined by X-ray and neutron diffraction methods ?

(10)

3. (a) Complete the following equation and identify X, Y, Z, Q, and R.

(i) $^{24}_{11}Na + ^{1}_{0}n \rightarrow ^{24}_{10}X + ^{1}_{1}Y$.

(ii) $Z + ^{4}_{2}Q \rightarrow ^{13}_{7}R + ^{1}_{0}n$. (3)

(*Or*)

(b) What is artificial transmutation ? Give the nuclear reactions induced by proton. (3)

(c) Explain the principle of G.M.Counter.

(7)

(*Or*)

(d) What is meant by Isotopic dilution technique ? Explain. (7)

(e) (i) Energy is released in nuclear fusion as well as nuclear fission. Explain why. (5)

(ii) Explain the nuclear fission is a chain reaction. (5)

(*Or*)

(f) (i) Explain the principle involved in hydrogen bomb. (5)

(ii) Write a note on atomic bomb. (5)

4. (a) What is meant by reductive elimination reaction ? (3)

(*Or*)

(b) What are organometallic compounds ? (3)

(c) Explain Dewar-Chatt approach to bonding in olefins. (7)

(*Or*)

(d) Discuss the general method for the preparation of organometallic compounds. (7)

(e) Explain the structure of ferrocene. (10)

(*Or*)

(f) Write a note on Zeigler - Natta polymerisation reaction. (10)

5. (a) What is van Arkel method of purification of titanium ? (3)

(*Or*)

(b) What are chief ores of magnesium ? (3)

(c) Name the important ores of molybdenum. How is the metal extracted ? (7)

(*Or*)

(d) How is tungsten extracted form its ore ?

(7)

(e) (i) Name the principal ores of copper. Explain electrolytic refining process for purification of copper. (5)

(ii) Describe the extraction of nickel by Mond's process. (5)

(*Or*)

(f) (i) Explain the manufacture of steel by Bessemer's Process.

(7)

(ii) Write a note on Tempering steel, Hardening steel and case hardening steel.

(3)

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RW-6598 **605103**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Chemistry (Week-End)

PHYSICAL CHEMISTRY - I

(NCBCS—1995–2001 Batch)

Time : 3 Hours Maximum : 100 Marks

Section A (5 × 20 = 100)

Answer **all** questions.

1. (a) Calculate the mean activity coefficient of Na_2SO_4 at a molality of 0.001 in aqueous solution at 25°C. (6)

(*Or*)

(b) Derive the Gibbs-Duhem equation and explain its significance (6)

(c) Derive Gibbs - Helmholtz equation and indicate its applications (7)

(*Or*)

(d) Give the statistical meaning of third law of thermodynamics. What are the apparent exceptions to this law?

(7)

(e) Define the term fugacity. How the graphical method used to determine the fugacity of gas ?

(7)

(*Or*)

(f) Discuss in detail about partial molal quantities.

(7)

2. (a) Explain the Born model of ion-solvent interactions.

(6)

(*Or*)

(b) Write a note on electrode potential

(6)

(c) Derive the Debye-Hückel-Onsager conductivity equation.

(7)

(*Or*)

(d) Discuss the applications of conductivity measurements.

(7)

(e) Derive the Butler-Volmer equation

(7)

(*Or*)

(f) Discuss the applications of EMF measurements

(7)

3. (a) Which of the functions are eigenfunctions of $\frac{d^2}{dx^2}$?

(i) $\sin 3x$. (6)

(ii) $5x^3$.

(*Or*)

(b) What are the basic postulates of quantum mechanics ?

(6)

(c) Write a note on the following :

(7)

(i) Wave particle duality

(ii) Uncertainty principle.

(*Or*)

(d) Derive an expression for the energy and wave function equations of a particle in a one-dimensional box.

(7)

(e) Explain the different types (any three) operators. (7)

(*Or*)

(f) Derive an expression for the energy of a particle in a three dimensional box. (7)

4. (a) Explain the various photophysical processes using Jablonski diagram. (6)

(*Or*)

(b) Explain the term Hydrated electron. (6)

(c) What are quantum yield ? Give the experimental determination of quantum yield. (7)

(*Or*)

(d) Explain the term (7)

(i) Radiolysis of water

(ii) Photosensitization.

(e) Discuss in detail about photochemical decomposition of carbonyl compounds. (7)

(*Or*)

(f) Explain the following : (7)

(i) Flash photolysis

(ii) Energy transfer mechanism.

5. (a) Distinguish between opposing, consecutive and parallel reactions. (6)

(*Or*)

(b) State and explain microscopic reversibility. (6)

(c) Indicate the important difference between simple collision theory and ARRT. (7)

(*Or*)

(d) Discuss in detail about kinetic isotopic effect. (7)

(e) Describe the theory of unimolecular reactions based on Lindemann's concept. (7)

(*Or*)

(f) Write notes on branched chain and explosive reactions. (7)

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RW-6599 **605104**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Chemistry (Week End)

INSTRUMENTAL METHODS OF ANALYSIS

(CBCS—1995–2001 Batch)

Time : 3 Hours Maximum : 100 Marks

Part A (5 × 20 = 100)

Answer **all** questions.

All questions carry equal marks

1. (a) Which structure feature may produce a bathochromic or a hypsochromic effect in an organic compound ? (3)

(*Or*)

(b) Predict the frequency shift of the carbonyl absorption in the following aldehydes. (1–3)

Cyclohexane aldehyde (1)

Benzaldehyde (2)

Salicylaldehyde (3)

(c) Describe in detail the effect of solvent in carbonyl absorption.

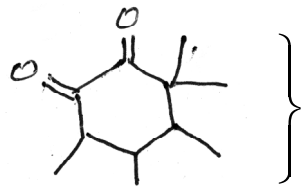
(7)

(*Or*)

(d) Write a note on charge transfer complexes.

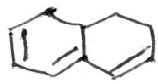
(7)

(e) (i) How will you confirm the presence of α - diketone system in the following steroid ?



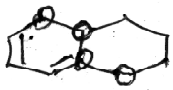
(4)

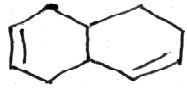
(ii) The following triene on partial hydrogenation gives three products which are separated by GLC. How UV spectroscopy and application of Woodward Fieser rules will help to identify the products?



$$\xrightarrow[\text{Pt}]{H_2(1\,eq)}$$ unknown structure

$C_{10}H_{14}$





(6)

(*Or*)

(f) (i) Discuss in detail the various factors which influence the vibrational frequency of a particular group. Give examples.

(7)

(ii) Ring strain in a lactone increases the C=O stretching frequency. Illustrate with examples. (3)

2. (a) Why is TMS used as an internal standard ?

(3)

(*Or*)

(b) Distinguish between g values of NMR and ESR.

(3)

(c) (i) The low natural abundance of ^{13}C (1.1%) is both an advantage and disadvantage Substantiate. (4)

(ii) Write a note on deshielding due to hydrogen bonding in NMR spectrum taking a specific example. (3)

(*Or*)

(d) State the principle of ^{13}CNMR. How will you distinguish between *cis* an *trans*-2-butene on the basis of ^{13}C NMR spectroscopy?

(7)

(e) Explain briefly the theory and application of ESR spectroscopy to inorganic compounds (10)

(*Or*)

(f) (i) What is meant by nuclear Overhauser effect? Explain briefly with examples.

(6)

(ii) What is FT-NMR? Explain the advantages of FT-NMR.

3. (a) What is meant by α-haloketone rule? Explain

(3)

(*Or*)

(b) What is TLC ? State its principle and important applications.

(3)

(c) Explain briefly the Mclafferty rearrangement with suitable example.

(7)

(*Or*)

(d) Explain the meta stable ion, base peak, parent ion and isotopic ion in mass spectroscopy with suitable examples.

(7)

(e) (i) State and explain the principle of gas chromatography. (4)

(ii) What is meant by HPLC? Mention its principle and important applications. (6)

(*Or*)

(f) What are positive and negative cotton effects? Explain their applications with suitable examples.

4. (a) State and explain the principle of colorimetry. (3)

(*Or*)

(b) State the principle of fluorimetry. Explain (3)

(c) What is meant by nephelometry? Explain briefly its principle and applications. (7)

(*Or*)

(d) Explain the basic principle and applications of flame photometry. (7)

(e) Discuss briefly the principle and applications of differential thermal analysis. (10)

(*Or*)

(f) State and explain the principle of DSC. Discuss its important applications. (10)

5. (a) What is meant by Coulometric analysis? State its principle. (3)

(*Or*)

(b) What are F-test and Q-test? Mention their importance. (3)

(c) What is meant by amperometry? Explain its principle and applications. (7)

(*Or*)

(d) What is stripping analysis? What are its two types? Explain the basic principles. (7)

(e) Discuss briefly the principle instrumentation and applications of electrogravimetry. (3)

(*Or*)

(f) (i) What are errors? What are its two types? (3)

(ii) Distinguish between Precision and Accuracy (3)

(iii) Explain the various methods to minimize the errors. (4)

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RW-6600 **605201**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Chemistry (Week–End)

ORGANIC CHEMISTRY—II

(NCBCS–1995–2001 Batch)

Time : 3 Hours Maximum : 100 Marks

(5 × 20 = 100)

Answer **all** questions.

1. (a) Give an example for Clemmension reduction.

(3)

Or

(b) Explain Birch reduction.

(c) Discuss oxidation reactions using DMSO and MnO_2.

(3.5 + 3.5)

Or

(d) Enumerate the applications of chromic acid in organic synthesis.

(7)

(e) Describe the mechanism of the following :

(i) Demzenov rearrangement,

(ii) Baeyer-Villiger rearrangement.

(5 + 5)

Or

(f) (i) Stevens rearrangement.

(ii) Cope rearrangement.

(5 + 5)

2. (a) Write short notes on racemisation. (3)

Or

(b) Explain Crams rule with suitable example.

(c) Explain Cahn-Prelog-Ingold nomenclature using bromochloroiodo methane as an example.

(7)

Or

(d) Discuss about enantiotopic and diastereotopic isomers.

(e) (i) Discuss about the optical activity of sulphur compounds.

(5)

(ii) Give an account on isomerism exhibited by carbon-carbon double bonds.

(5)

Or

(f) Define the following terms :

(i) Chiral.

(ii) Diastereomers.

(iii) Stereoselective.

(iv) Prochiral.

(v) Resolution.

(2 × 5)

3. (a) Explain conformation with example. (3)

Or

(b) Give an example for *cis* elimination. (3)

(c) Give an account on formation and stability of free radicals.

(7)

Or

(d) Discuss about the mechanism of chlorination of methane.

(7)

(e) Explain the following name reactions :

(i) Gomberg reaction.

(ii) Ullman reaction.

(iii) Hunsdiecker reaction.

(iv) Pschorr reaction.

(2. 5 × 4)

Or

(f) (i) Explain Free radical polymerization and addition reactions.

(2 × 2.5)

(ii) Discuss neighbouring group participation.

(5)

4. (a) Define the term Quantum yield.. (3)

Or

(b) Explain Photoreduction reaction.

(c) Give an account on photochemical reactions of olefins.

(7)

Or

(d) Explain Di-Pi methane rearrangement.

(e) Discuss in detail about sigmatropic reactions.

(10)

Or

(f) Give an account on Woodward-Hoffman rules in detail.

(10)

5. (a) Draw the structures of oesterone. (3)

Or

(b) Write the structure of product when ergosterol undergoes Oppenauer oxidation.

(c) Describe the synthesis of cholestrol from cholestanol.

(7)

Or

(d) Describe the synthesis of progesterone from cholestrol.

(e) Establish the structure of testosterone.

(10)

Or

(f) Describe the Johnsons method of synthesis of (+) equilenin.

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RW-6601 **605202**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Chemistry—Week End

INORGANIC CHEMISTRY—II

(NCBCS 1995-2001 Batch)

Time : 3 Hours Maximum : 100 Marks

(5 × 20 = 100)

Answer **all** questions.

All questions carry equal marks.

1. (a) Why is LF SE minimum in lanthanide complexes.

Or

(b) How will you extract a mixture of Lanthanides from Monazite sand ?

(c) Explain the following :

(i) Complex formation tendency of Lanthanide cations.

(ii) Reducing property of Lanthanide ions.

Or

(d) Explain the principles involved in the separation of Lanthanide by

(i) Solvent extraction and

(ii) Ion-exchange methods.

(e) Justify the position of actinides in the periodic table.

Or

(f) What is Lanthanide contraction ? How would you account for it ? Discuss its consequences.

2. (a) Distinguish between stepwise stability constant and overall stability constant.

Or

(b) $\left[Ni(CN)_4\right]^{2-}$ is very stable and the $t_{\frac{1}{2}} = 3\ OS$ for the following reaction

$$\left[Ni(CN)_4\right]^{2-} + 4^{*}CN^{-} \rightarrow \left[Ni^{*}(CN)_4\right]^{2-} + 4\,CN^{-}$$

comment on this.

(c) Explain the mechanism of acid hydrolysis reaction.

Or

(d) Discuss the mechanism of a equation of cis- and trans - $\left[Co(en)_2\, cl(OH)\right]^{+}$ complexes.

(e) What do you understand by nucleophilic substitution reactions for octahedral complexes ? Discuss the S_N1 and S_N2 mechanisms for the base hydrolysis of octahedral complexes.

Or

(f) Describe briefly about the geometrical isomerisation in square planar complex giving a general mechanism.

3. (a) Give two examples for oxidation reduction reactions occuring through the transfer of atoms of groups.

Or

(b) What is the essential condition for an electron transfer reaction to take place by inner sphere mechanism ?

(c) What do you understand by the term " Template Effect". Explain with two examples

Or

(d) Explain with examples, the adiabatic and non-adiabatic processes of electron transfer reactions.

(e) State and explain-Marcus-Hush theory of electron transfer reactions. Discuss the type of mechanism involved.

Or

(f) What are complementary and non-complementary electron transfer reactions ? How will you distinguish them ? Explain the mechanism of these reactions with suitable examples.

4. (a) Explain why square planar N*i* (II) organometallic complexes are stable although they do not obey 18 electron rule.

Or

(b) How is $V(Co)_6$ prepared ?

(c) Discuss the acidity and basicity of carbonyl hydrides.

Or

(d) State and explain the uses of Zeigler-Natta catalysts.

(e) Write any two methods of preparing Fe $(Co)_5$. Discuss its structure and bonding.

Or

(f) (i) Explain how acetic acid is manufactured using rhodium catalyst.

(ii) What is the essential condition for an organometallic compound to act as a catalyst ?

5. (a) What are Zeolites ? Give their general formula.

Or

(b) What is metal cluster ? Illustrate your answer with two examples.

(c) Describe the structure of diborane based on MOT.

Or

(d) Draw and explain the structure and bonding in the eclipsed and staggered conformation of any complexanion.

(e) Give any one method of preparing phosphonitrilic polymers and describe the structure of the trimer.

Or

(f) Draw and discuss the structure of

(i) Poly phosphoritic chloride.

(ii) Hexachloro cyclo triphasphazine and

(iii) Octachloro cyclo tetra phosphazine.

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RW-6602 **605203**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Chemistry (Week-End)

PHYSICAL CHEMISTRY–II

(NCBCS-1995–2001 Batch]

Time : 3 Hours Maximum : 100 Marks

(5 × 20 = 100)

Answer **all** questions.

All questions carry equal marks.

1. (a) Define thermodynamic probability of a macrostate. (3)

(*Or*)

(b) Define partition function and give its significance. (3)

(c) Explain Translational, vibrational and rotational partition function. (7)

(*Or*)

(d) Explain Onsager reciprocal relations and discuss non-equilibrium thermodynamics. (7)

(e) Utilize the partition function to evaluate Internal energy, entropy, heat capacity and work function. (10)

(*Or*)

(f) Discuss the theories of heat capacity of solids. (10)

2. (a) Explain the concept of the Dual nature of electron using de Brogli's equation. (3)

(*Or*)

(b) Discuss Heisenbergs Uncertainty principle. (3)

(c) Discuss the rigid rotor and why it has two quantum number. What is their meaning. (7)

(*Or*)

(d) Discuss the orthogonality of ψ_o and ψ_- for a harmonic oscillator and how to calculate the uncertainty in momentum for a harmonic oscillator.

(7)

(e) (i) In case of hydrogen molecule discuss Uo theory and LACO for obtaining symmetric and anti symmetric energy states.

(5)

(ii) Discuss the stability of hydrogen molecule ion

(5)

(*Or*)

(f) Discuss in detail bonding and Anti bonding orbitals. (10)

3. (a) Taking a cube or a water molecule classify the symmetry operation. (3)

(*Or*)

(b) Define a group and discuss the properties using a water molecule. (3)

(c) Construct the multiplication table for the group C_{3v}. (7)

(*Or*)

(d) Construct molecular orbitals of proper symmetry for the formaldehyde molecule, $H_2C=0$, symmetry group C_{2v}. (7)

(e) State and explain great orthoganality theorem and discuss how to construct MO for NH_3 using LACO showing the resulting MO_3 belong to a reducible representation of the group C_{3v}. (10)

(f) Discuss how to apply group theory to IR and Raman spectra of H_2O and NH_3 molecules. (10)

4. (a) With suitable example explain the theory of homogeneous catalysis. (3)

(*Or*)

(b) Explain the effect of temperature on reaction rate. (3)

(c) Predict the effect of the increase in ionic strength on the rate constant for each of the following reactions. (7)

(i) $H_2O_2 + H^+ + Br^-$.

(ii) $C_{12}H_{22}O_{11} + OH^-$.

(iii) $S_2O_8^{2-} + I^-$.

(*Or*)

(d) Discuss Zucker-Hammett hypothesis and give its significance. (7)

(e) Derive double sphere model equation for the effect of dielectric constant of the reaction. (10)

(*Or*)

(f) Discuss Hammett equation and give its significance. Explain σ and ρ. (10)

5. (a) Why should the Langmuir adsorption isotherm be more reliable, at high gas pressures, for chemisorption than for physical adsorption.

(3)

(*Or*)

(b) An excess pressure of 364 Pa is required to produce a hemispherical bubble at the end of a Capillary tube of 0.300 mm diameter immersed in acetone. Calculate γ. (3)

(c) Derive Gibbs equation and give its significance.

(7)

(*Or*)

(d) Describe the role of both the inner and outer portion of the nice lle in the action of soap.

(7)

(e) Derive B.E.T adsorption isotherm and give its significance. (10)

(*Or*)

(f) (i) Discuss the effect of temperature on adsorption. (5)

(ii) Write a note on electrical phenomena at the interface. (5)

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RW-6603 **605204**

M.Sc. DEGREE EXAMINATION, APRIL 2011

Chemistry (Week-End)

APPLIED CHEMISTRY

(NCBCS—1995-2001 Batch)

Time : 3 Hours Maximum : 100 Marks

(5 × 20 = 100)

Answer **all** questions.

All questions carry equal marks.

1. (a) What is mono and bipolar electrodes ? What are the roles of Diaphragms and separators in an electrolytic cell. Explain the water electrolysis.

(20)

(*Or*)

(b) Write the chemical and electrochemical methods of manufacturing of Sodium and Potassium chlorates.

(20)

2. (a) What are the different types of primary batteries are available ? What are its applications ? What are the various components in a battery explain the role of each component ?

(20)

(*Or*)

(b) Write in detail about : (Each 5 marks)

(i) Lithium batteries.

(ii) Nickel metal hydride.

(iii) Nickel-Iron.

(iv) Nickel-cadmium.

3. (a) What are the different types of Corrosion. (10)

(b) What is pourbaix diagram. Explain with an example. (10)

(*Or*)

(c) What are the electrochemical methods of corrosion protection. (10)

(d) What are different methods of Corrosion Monitoring. (10)

4. (a) Write the mechanism of electrodeposition from a simple salt and complex salt. (10)

(*Or*)

(b) Write the short notes on the following :

(Each 5 marks)

(i) Electropolishing.

(ii) Phosphating and Chromating.

(iii) Electroforming.

(iv) Electroless plating.

5. (a) How polymers are classified ? (10)

(b) Mention the biomedical applications of polymers. (5)

(c) Define Glass Transition Temperature. (5)

(*Or*)

(c) What are Conducting Polymers. What are the different Methods of Preparing conducting Polymers. (10)

(e) How Polymers are characterised ? (10)

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